

Empirical Semantic Information Ladder

semantic observability increases
computational cost increases

Evidence level	Mean MI	Rel. SHAP	Incremental F	Mean cost	Main uncertainty removed
Execution	0.420	--	+0.314	30.305 ms	tested behavioral outcome
Data flow	0.002	0.17	+0.020	0.064 ms	value propagation
Identifier roles	0.009	0.23	+0.011	0.137 ms	value responsibility
Operator semantics	0.030	0.66	+0.042	0.088 ms	local computation
Control flow	0.020	0.36	-0.003	0.287 ms	possible execution paths
Program structure	0.014	0.16	+0.044	3.136 ms	organizational form
Syntax	0.036	1.00	+0.090	1.847 ms	grammatical possibility
Lexical evidence	0.033	0.68	+0.362	0.086 ms	symbol correspondence

evidence-supported confidence increases
semantic uncertainty decreases